Algorithm Description Document

[ALGORITHM NAME — e.g., "SDA Tracking Layer GNC: Threat Detection and Attitude Control"]

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Prepared by:	Navier AI STELLA Agent
Reviewed by:	[MANUAL: Engineer name]

NOTICE: This document was generated automatically by the STELLA post-processing agent from simulation telemetry bundle *[AUTO: bundle ID]*. All numerical values are derived from simulation outputs. Engineer review required before use as a program deliverable.

1 Document Control

1.1 Revision History

Rev	Date	Description	Author
A	[AUTO: date]	Initial release — auto-generated from STELLA bundle	STELLA Agent

1.2 Applicable Documents

ID	Title	Version
AD-01	STELLA Framework Architecture Description	current
AD-02	SDA PWSA Interface Control Document	[MANUAL]
AD-03	EGM2008 Gravitational Model	2008
AD-04	[MANUAL: additional program docs]	

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2 Introduction

2.1 Purpose

This document describes the guidance, navigation, and control (GNC) algorithms implemented in the STELLA simulation framework for the *[AUTO: vehicle/mission name]* mission. It provides the mathematical basis, input/output interface definition, mode logic, and simulation-derived performance characterization required for algorithm review and program documentation.

2.2 Scope

This ADD covers the following algorithm functions:

- [AUTO: list modes/algorithms from telemetry, e.g., "Detumble", "NadirPointing", "SlewToTrack", "Tracking", "ReturnToNadir"]

The following are explicitly *out of scope*: [MANUAL]

2.3 Definitions and Acronyms

Term	**Definition**
ADD	Algorithm Description Document
ECI	Earth-Centered Inertial frame
ECEF	Earth-Centered Earth-Fixed frame
EKF	Extended Kalman Filter
FSW	Flight Software
GNC	Guidance, Navigation, and Control
OPIR	Overhead Persistent Infrared
PWSA	Proliferated Warfighter Space Architecture
SDA	Space Development Agency
SITL	Software-in-the-Loop
SNR	Signal-to-Noise Ratio
[AUTO: add from telemetry bundle metadata]	

3 System Overview

3.1 Vehicle Configuration

Parameter	Value
Vehicle ID	[AUTO]
Orbit Altitude	[AUTO] km
Inclination	[AUTO]°
Dry Mass	[AUTO] kg
Moments of Inertia	Ixx=[AUTO], Iyy=[AUTO], Izz=[AUTO] $kg \cdot m^2$
Actuators	[AUTO: e.g., reaction wheels, thrusters]
Sensors	[AUTO: e.g., star tracker, GPS, IMU]
Payload FOV	[AUTO]° half-angle

3.2 GNC Architecture

The GNC system is structured as a finite state machine with the following modes:

[AUTO: mode transition diagram — generate from telemetry mode_id column]

Figure 1: GNC Mode State Machine

4 Navigation Algorithms

4.1 State Estimation: Extended Kalman Filter

4.1.1 State Vector

The navigation filter maintains the following state vector:

$$\boldsymbol{x} = \left[\boldsymbol{r}^T, \boldsymbol{v}^T, \boldsymbol{q}^T, \boldsymbol{b}_\omega^T, \boldsymbol{b}_a^T\right]^T$$

where $\boldsymbol{r} \in \mathbb{R}^3$ is position in ECI (m), $\boldsymbol{v} \in \mathbb{R}^3$ is velocity (m/s), $\boldsymbol{q} \in \mathbb{R}^4$ is the attitude quaternion (body-to-ECI), $\boldsymbol{b}_\omega \in \mathbb{R}^3$ is gyro bias (rad/s), and $\boldsymbol{b}_a \in \mathbb{R}^3$ is accelerometer bias (m/s^2).

4.1.2 Process Model

4.1.3 Measurement Models

Star Tracker:

$$\boldsymbol{z}_{\mathrm{ST}} = \boldsymbol{q}_{\mathrm{meas}} + \boldsymbol{\nu}_{\mathrm{ST}}, \quad \boldsymbol{\nu}_{\mathrm{ST}} \sim \mathcal{N}(0, \boldsymbol{R}_{\mathrm{ST}})$$

GPS:

$$\boldsymbol{z}_{\mathrm{GPS}} = \boldsymbol{r}_{\mathrm{true}} + \boldsymbol{\nu}_{\mathrm{GPS}}, \quad \boldsymbol{\nu}_{\mathrm{GPS}} \sim \mathcal{N}(0, \boldsymbol{R}_{\mathrm{GPS}})$$

IMU:

$$\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_{\mathrm{true}} + \boldsymbol{b}_\omega + \boldsymbol{\nu}_\omega$$

4.1.4 Filter Performance (Simulation Results)

[AUTO: position error norm vs. time plot]

Figure 2: Navigation Position Error — 3σ bounds from simulation

Metric	Mean	3σ	Requirement
Position error (m)	[AUTO]	[AUTO]	[MANUAL]
Velocity error (m/s)	[AUTO]	[AUTO]	[MANUAL]
Attitude error (deg)	[AUTO]	[AUTO]	[MANUAL]

5 Attitude Control Algorithms

5.1 Mode: Detumble

5.1.1 Algorithm Description

B-dot control law for initial detumble following separation or anomaly recovery:

$$\boldsymbol{\tau}_{\text{cmd}} = -k_{\text{bdot}} \cdot \dot{\boldsymbol{B}}_{\text{body}}$$

where $\boldsymbol{B}_{\text{body}}$ is the measured magnetic field vector in body frame and k_{bdot} is the detumble gain.

5.1.2 Parameters

Parameter	Value	Units
k_{bdot}	[AUTO]	$\text{N} \cdot \text{m} \cdot \text{s}/\text{T}^2$
Exit condition: $\lvert\boldsymbol{\omega}\rvert$	[AUTO]	rad/s

5.1.3 Simulation Results

[AUTO: angular rate vs. time during detumble phase]

Figure 3: Angular Rate Convergence During Detumble

5.2 Mode: Nadir Pointing

5.2.1 Algorithm Description

PD attitude controller with nadir-pointing reference quaternion:

$$\boldsymbol{\tau}_{\text{cmd}} = -K_p \boldsymbol{q}_{\text{err,vec}} - K_d \boldsymbol{\omega}_{\text{err}}$$

where $\boldsymbol{q}_{\text{err}}$ is the error quaternion between current and desired attitude, and $\boldsymbol{\omega}_{\text{err}}$ is the angular rate error.

5.2.2 Parameters

Parameter	Value	Units
K_p	[AUTO]	$\text{N} \cdot \text{m}/\text{rad}$
K_d	[AUTO]	$\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}$

Pointing accuracy (3σ)	[AUTO]	deg

5.3 Mode: Slew to Track

5.4 Mode: Threat Tracking

5.4.1 Detection Logic

Threat detection is performed by checking whether the threat position vector, transformed into sensor frame, falls within the sensor field of view:

$$\text{detect} = \begin{cases} \text{true if } \angle\left(\hat{\boldsymbol{r}}^{\text{sensor}}_{\text{threat}}, \hat{\boldsymbol{z}}_{\text{boresight}}\right) \leq \theta_{\text{FOV}} \text{AND SNR} > \text{SNR}_{\text{threshold}} \\ \text{false otherwise} \end{cases}$$

where $\theta_{\text{FOV}} = [\text{AUTO}]^\circ$ is the sensor half-angle.

5.4.2 Track Filter

5.4.3 Tracking Performance

Metric	Mean	3σ	Requirement
Track position error (m)	[AUTO]	[AUTO]	[MANUAL]
Track velocity error (m/s)	[AUTO]	[AUTO]	[MANUAL]
Time to weapons quality (s)	[AUTO]	—	[MANUAL]

[AUTO: tracking error vs. time with weapons-quality threshold line]

Figure 4: Threat Track Error — Time to Fire-Control Quality

6 Guidance Algorithms

6.1 Proportional Navigation

6.1.1 Algorithm Description

Pure proportional navigation guidance law:

$$\boldsymbol{a}_{\text{cmd}} = N' \cdot \boldsymbol{V}_c \cdot \boldsymbol{\Omega}_{\text{LOS}}$$

where N' is the effective navigation ratio, $\boldsymbol{V}_c$ is the closing velocity, and $\boldsymbol{\Omega}_{\text{LOS}}$ is the LOS rotation rate vector.

6.1.2 Flight Phase Parameters

Phase	Duration	N'	Notes
Boost	[AUTO] s	[AUTO]	
Midcourse	[AUTO] s	[AUTO]	
Terminal	[AUTO] s	[AUTO]	

6.1.3 Engagement Performance

Metric	Value
INT-1 miss distance	[AUTO] m — [AUTO: HIT/MISS]
INT-1 terminal phase duration	[AUTO] s
BDA delay	[AUTO] s
INT-2 miss distance	[AUTO] m — [AUTO: HIT/MISS]
Total engagement duration	[AUTO] s

7 Mission Timeline and Mode Transitions

[AUTO: swimlane timeline — each vehicle as a row, mode phases colored by state]

Figure 5: GNC Mode Transition Timeline — All Vehicles

T (s)	Vehicle	Mode Transition	Notes
[AUTO]	[AUTO]	[AUTO]	

8 Error Budget and Sensitivity Analysis

8.1 Navigation Error Contributions

Error Source	1σ Contribution	Notes
Star tracker noise	[AUTO]	
GPS measurement noise	[AUTO]	
IMU gyro noise	[AUTO]	
IMU accel noise	[AUTO]	
Gravity model error	[MANUAL]	EGM2008 assumed
Atmospheric drag (LEO)	[MANUAL]	
RSS Total	**[AUTO]**	

8.2 Injected Fault Analysis

Fault	Magnitude	Observed Effect
[AUTO: e.g., "Nav bias — INT-1"]	[AUTO]	[AUTO]

9 Verification and Validation

9.1 Simulation Configuration

Parameter	Value
STELLA version	[AUTO]
Physics rate	[AUTO] Hz
FSW rate	[AUTO] Hz
Simulation duration	[AUTO] s
Number of vehicles	[AUTO]
Scenario epoch	[AUTO]
Bundle ID	[AUTO]
Checksum	[AUTO]

9.2 Test Cases

ID	Description	Expected	Result
TC-01	Detumble convergence within time limit	[AUTO] s	[AUTO: PASS/FAIL]
TC-02	Nadir pointing accuracy	[AUTO] deg	[AUTO: PASS/FAIL]
TC-03	Threat detection latency	[AUTO] s	[AUTO: PASS/FAIL]
TC-04	Fire-control quality track acquisition	[AUTO] s	[AUTO: PASS/FAIL]
TC-05	Custody handoff continuity	No gap	[AUTO: PASS/FAIL]
TC-06	INT-2 intercept (post-BDA)	Miss < [AUTO] m	[AUTO: PASS/FAIL]

10 Known Limitations and Open Items

ID	Description	Status
LIM-01	Single-run simulation — Monte Carlo analysis pending	Open
LIM-02	Spherical Earth model — WGS84 ellipsoid not yet implemented	Open
LIM-03	Atmospheric drag model: exponential atmosphere, not NRLMSISE-00	Open
[AUTO]	[AUTO: flag any NaN, overflow, or anomaly detected in telemetry]	Open

11 References

1. Vallado, D.A., *Fundamentals of Astrodynamics and Applications*, 4th ed., Microcosm Press, 2013.
2. Montenbruck, O. and Gill, E., *Satellite Orbits*, Springer, 2000.
3. Wertz, J.R., *Spacecraft Attitude Determination and Control*, Kluwer, 1978.
4. EGM2008: Pavlis, N.K. et al., "The development and evaluation of the Earth Gravitational Model 2008," *JGR Solid Earth*, 2012.
5. [MANUAL: program-specific ICDs, SDA interface documents]

12 Appendix A: Telemetry Channel Definitions

Channel	Units	Description	Source
[AUTO]	[AUTO]	[AUTO]	STELLA telemetry

13 Appendix B: Full Telemetry Plots